Trust

In-context trust-learning under a reputation vs. accuracy conflict

Question: when a source's STATED REPUTATION conflicts with its OBSERVED ACCURACY,

does a frozen-weight LLM learn to trust the accurate source — and how fast vs. a rational learner?

Model: Qwen/Qwen3-8B (hf) | Source A = reputable but NOISIER, Source B = unestablished but ACCURATE

Accuracy gaps swept: sigma_A/sigma_B = [1.5, 2.0, 3.0, 5.0] (sigma_B=12.0) | Reputation strengths: ['faint', 'moderate', 'strong']

M=3 companies/round, T=32 rounds, ~20 games/condition | info-ladder: ['lean', 'self_history', 'summary', 'full']

21888 per-(round,company) observations | 2026-06-16

Setup overview. Reputation is verbal track-record framing only; it never updates and is decoupled from reward.

Method — per-round in-context loop

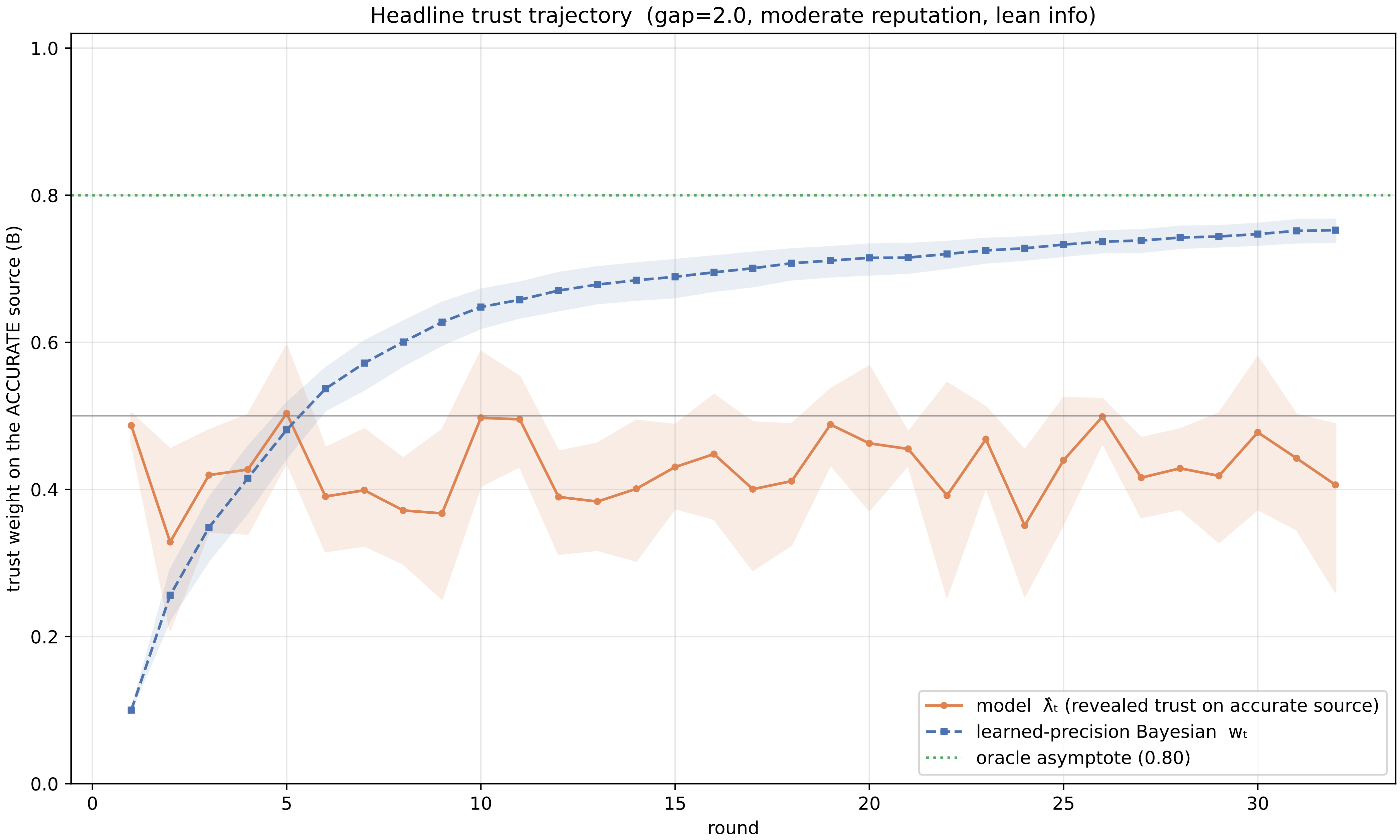
1. Build round-t prompt: persistent reputation header + RAW recap of past rounds (advisor record only) + this round's
2. Model outputs a numeric estimate per company (structured output) — the estimate IS the recommendation
3. Reveal the true value for each company -> accuracy evidence accumulates against the stale reputation
4. Extend history; proceed to round t+1 (one API call per round, T calls per game)

Reputation vs. accuracy CONFLICT: the header says A is trustworthy, but B's revealed errors are smaller every round.

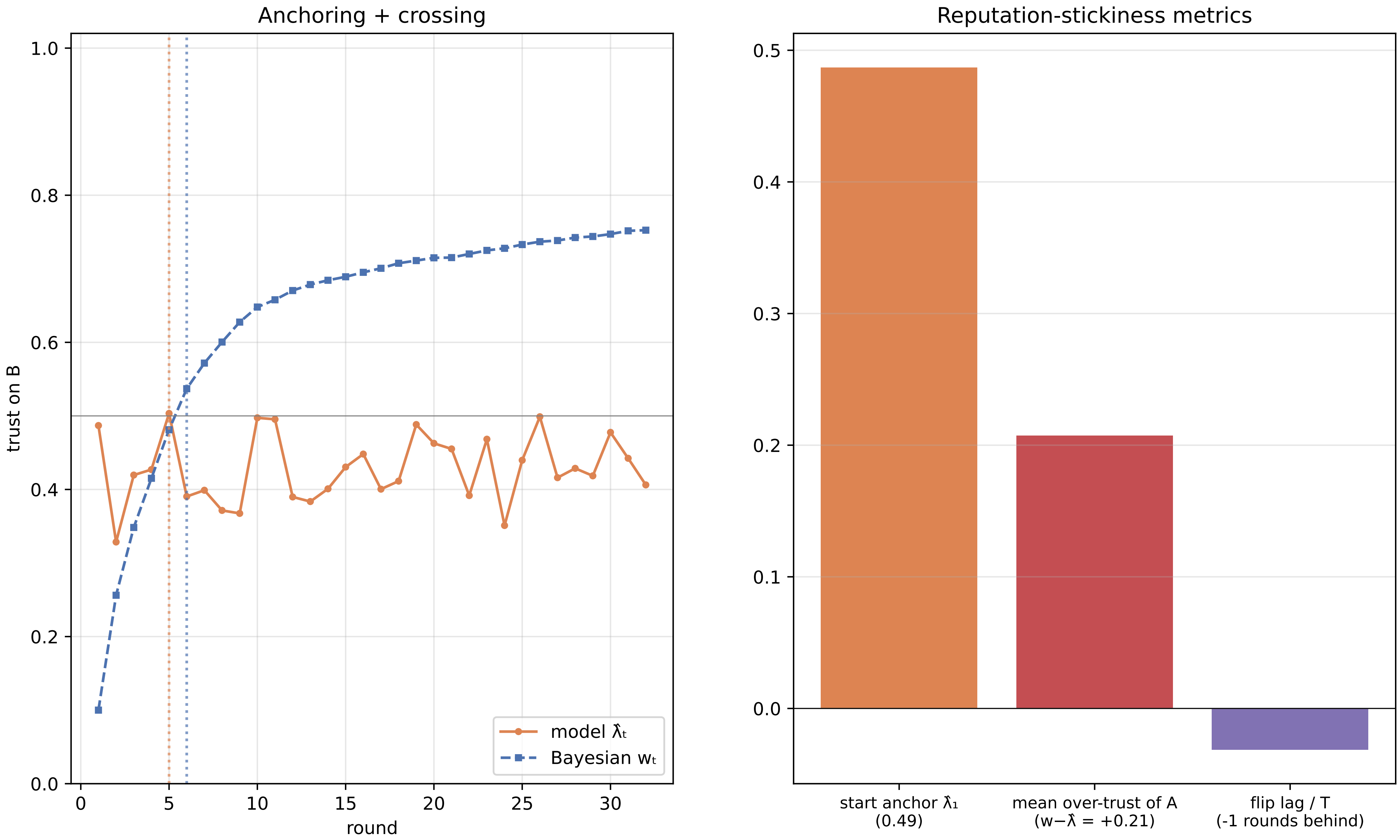
Normative baseline (LEARNED precision): a Bayesian that does NOT know the noise levels, starts from a reputation-derived
prior favoring A, and updates each source's precision from realized errors: pi_hat = (nu0 + n) / (nu0*sig0^2 + S).
Trust on B: w = pi_hat_B / (pi_hat_A + pi_hat_B) -> migrates A->B as evidence accrues (oracle asymptote uses the TRUE s

Model trust readout: per round t, regress model_est ~ bA*a + bB*b across companies x games; lambda_hat_t = bB/(bA+bB).

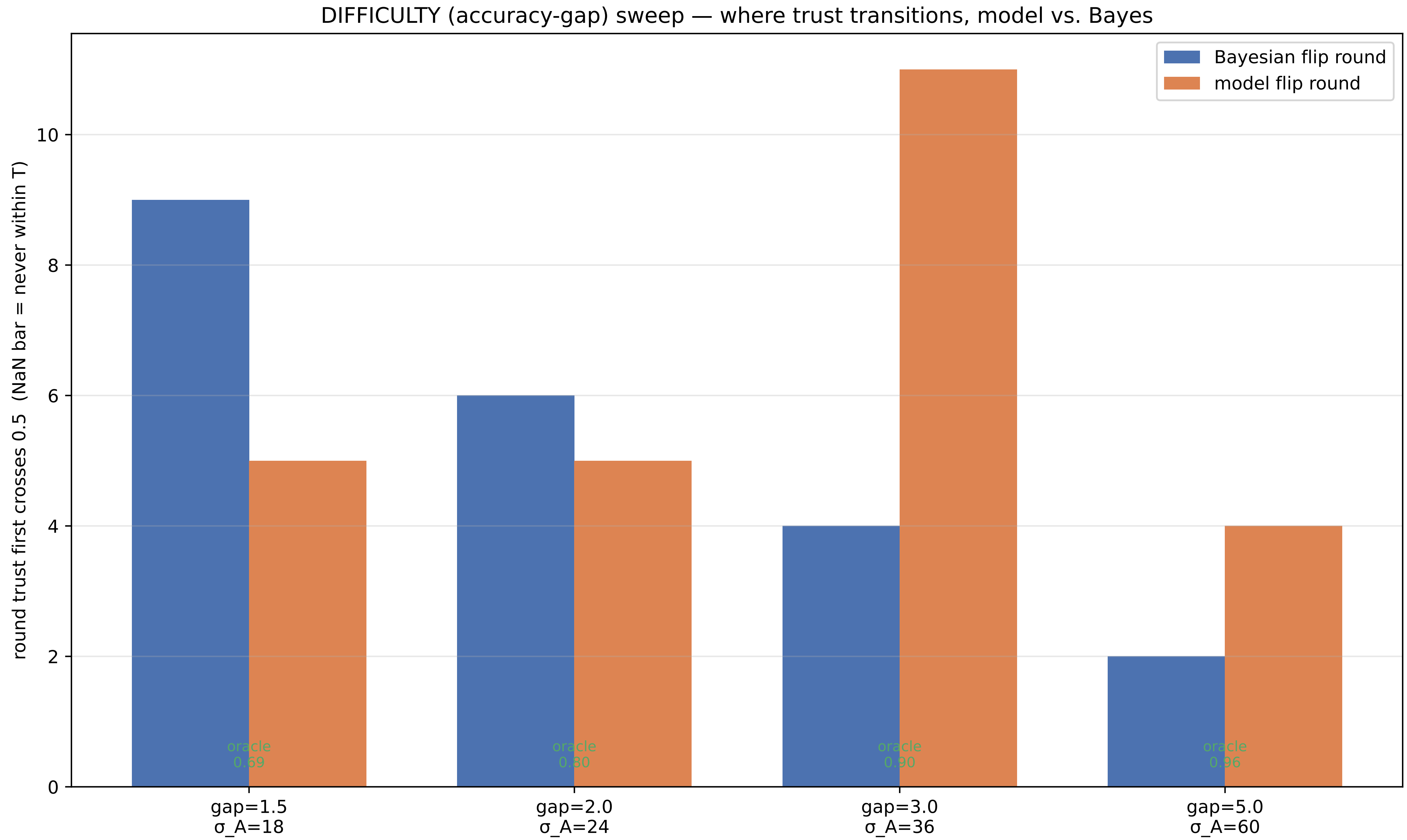
How one game runs and how the model's revealed trust (lambda_hat) and the rational trust (w) are computed.



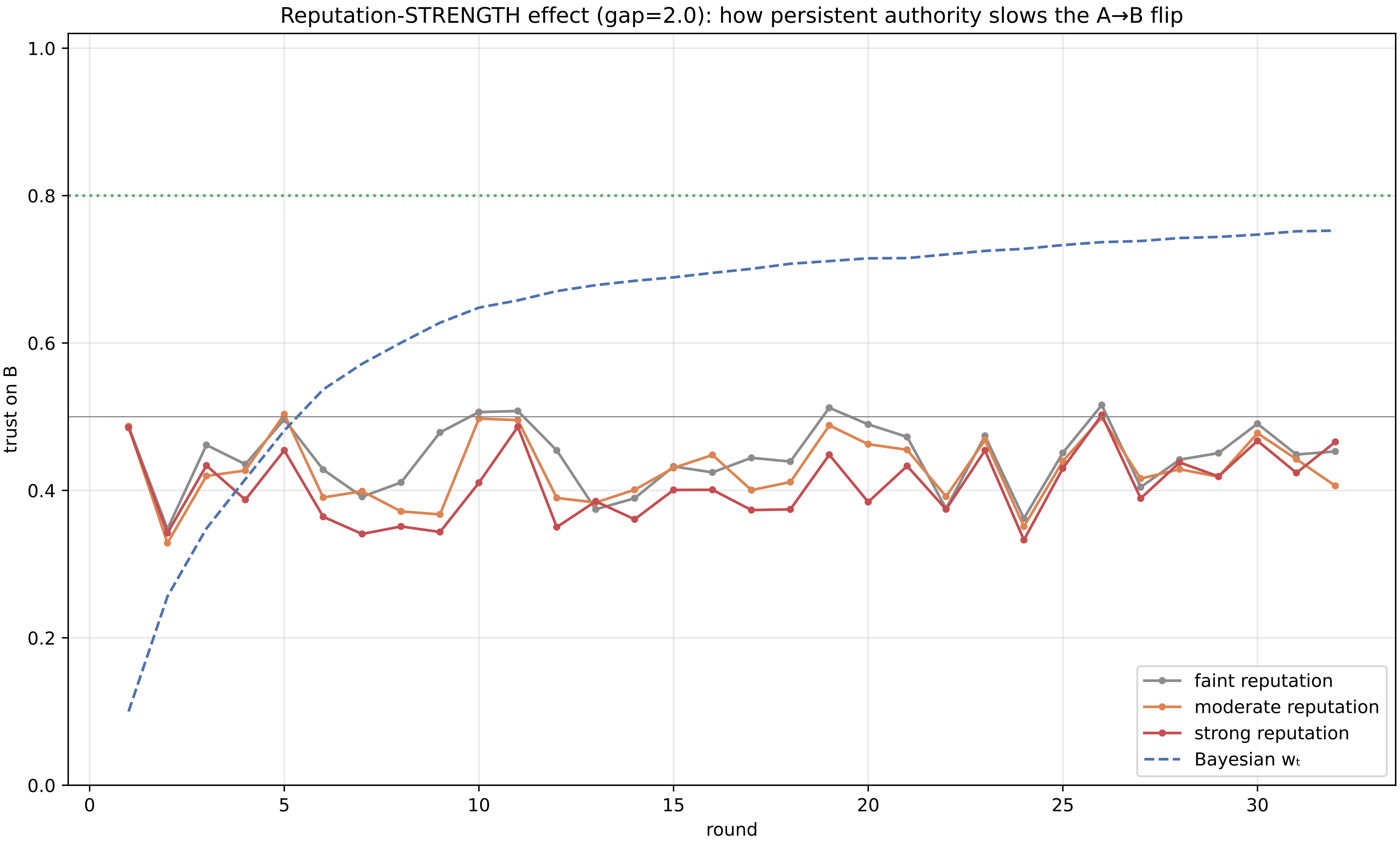
Both curves start anchored on the reputable-but-wrong source (low) and should climb toward the oracle. Model below Bayesian = reputation-stickiness; shaded = 95% bootstrap CI.



Stickiness = the model starts more anchored on A than warranted (left) and trails the rational A→B migration (positive over-trust, rounds-behind on the right).

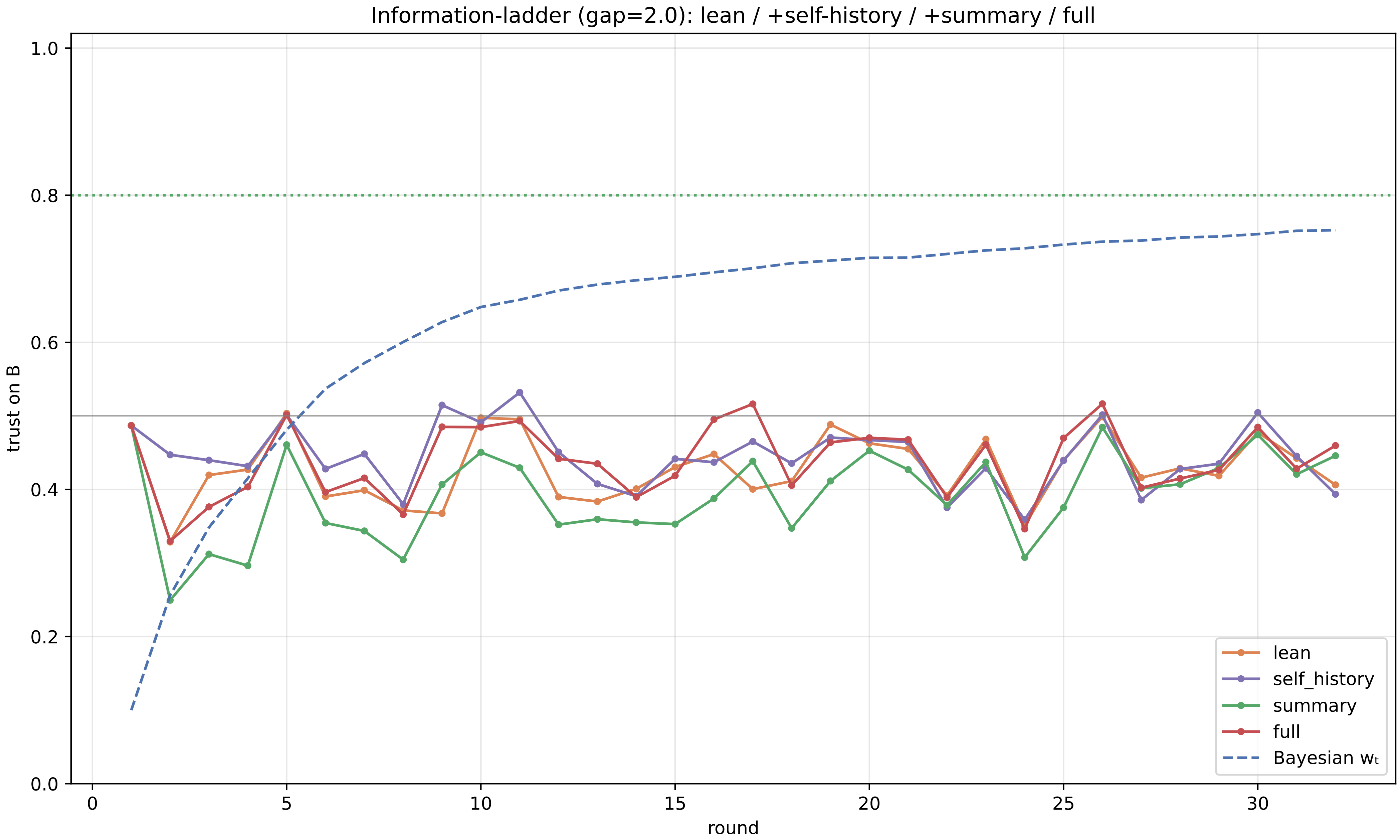


THE key plot: as the accuracy gap widens (easier to tell sources apart) both flip sooner. Taller model bars than Bayes = the model needs more evidence to overcome reputation.

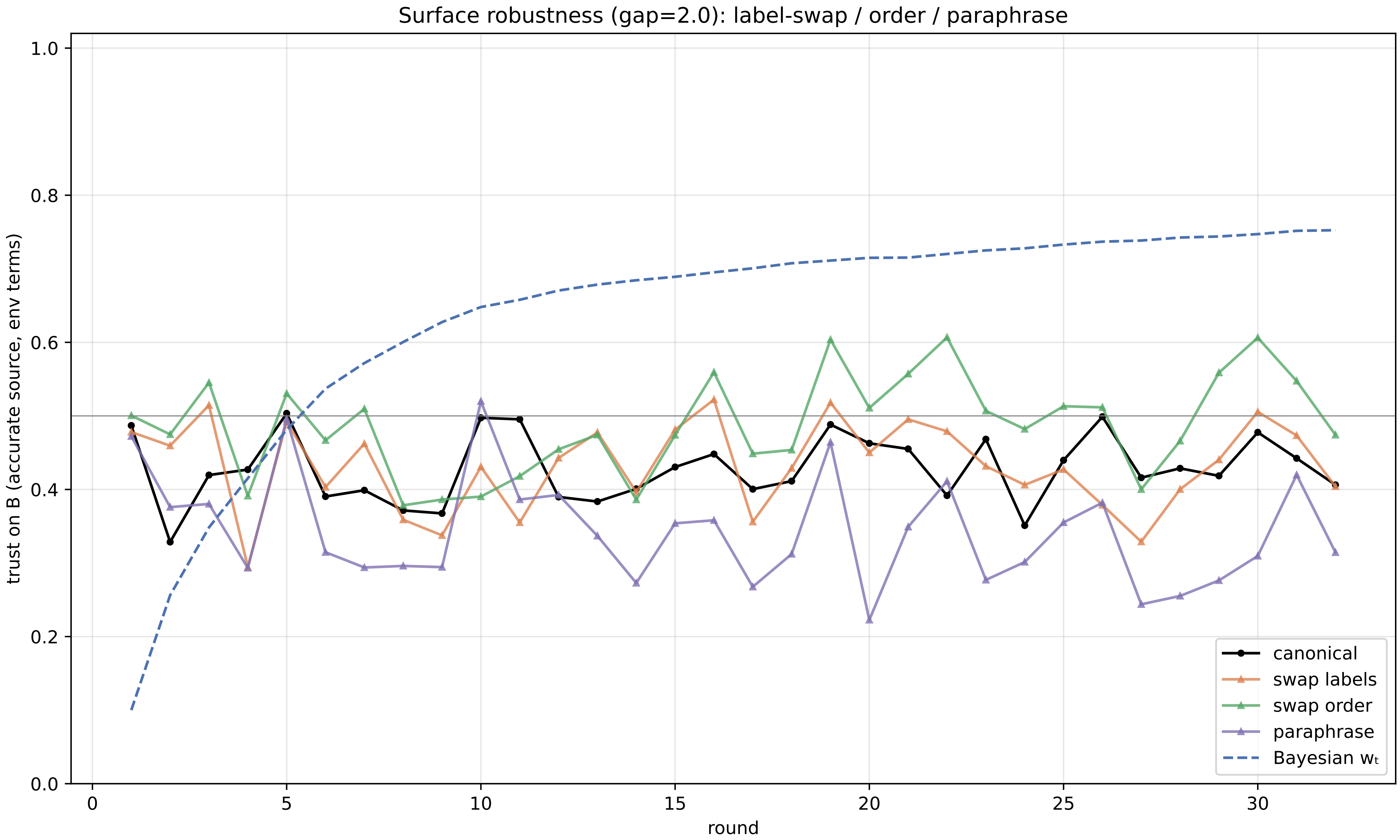


More persistently asserted reputation (faint→moderate→strong) should push the model's trajectory lower / later.

Gap between green (justified) and orange (model) at the end = residual deference to reputation that never resolves.



Localizes the bottleneck: if +summary (handed reliability) beats lean, it's an INFERENCE limit; if even full fails, it's DEFERENCE; if +self-history shifts it, anchoring.



Trajectories should overlap — the A→B migration shouldn't depend on which letter is reputable, the listing order, or wording.

Summary — flip round (model vs Bayes), lag, final trust, verdict

condition	gap	rep	info	flip model	flip Bayes	lag	final $\hat{\lambda}$	oracle	verdict
gap1.5_moderate_lea	1.5	moderate	lean	5	9	-4	0.43	0.69	learns→B
gap2.0_moderate_lea	2.0	moderate	lean	5	6	-1	0.44	0.80	learns→B
gap3.0_moderate_lea	3.0	moderate	lean	11	4	+7	0.48	0.90	sticky/slow
gap5.0_moderate_lea	5.0	moderate	lean	4	2	+2	0.58	0.96	learns→B
gap2.0_faint_lean	2.0	faint	lean	10	6	+4	0.46	0.80	sticky/slow
gap2.0_strong_lean	2.0	strong	lean	26	6	+20	0.45	0.80	sticky/slow
gap2.0_moderate_self_h	2.0	moderate	self_history	5	6	-1	0.45	0.80	learns→B
gap2.0_moderate_sumi	2.0	moderate	summary	—	6	—	0.45	0.80	no flip in T
gap2.0_moderate_fu	2.0	moderate	full	5	6	-1	0.46	0.80	learns→B
gap2.0_robust_swapla	2.0	moderate	lean	3	6	-3	0.46	0.80	learns→B
gap2.0_robust_orde	2.0	moderate	lean	1	6	-5	0.54	0.80	learns→B
gap2.0_robust_paraph	2.0	moderate	lean	10	6	+4	0.35	0.80	sticky/slow
gap2.0_moderate_no_ad	2.0	moderate	no_advisor	-	-	-	-	-	RMSE=129 (necessity)

lag = model flip round minus Bayesian flip round (positive = behind). final $\hat{\lambda}$ vs oracle shows residual deference.

Takeaways

- Does it learn to trust the accurate source? YES — at gap=2.0 the model's revealed trust crosses 0.5 at round 5 (Bayes: 6).
- Reputation-stickiness: starts anchored at $\hat{\lambda}_1$=0.49 (prior favors A) and over-trusts A by +0.21 on average vs. the rational curve; flip lag = -1 rounds.
- Where the transition sits vs. Bayes across gaps (model vs Bayes flip round): gap1.5:5/9, gap2.0:5/6, gap3.0:11/4, gap5.0:4/2.
- Asymptote: final $\hat{\lambda}$=0.44 vs accuracy-justified 0.80 — residual deference remains.
- Inference vs. deference: compare lean vs. +summary on S8 — if handing the model running accuracy helps, the bottleneck is inference; if not, it's deference.
- Sanity: no-advisor RMSE=129 (advisors must add info); comprehension correct in 100% of probes.
- Caveats: scoped run (~20 games/condition → wider CIs than the 1000-game target); thinking disabled; single model; synthetic Gaussian world.

Headline conclusions; see the summary table (S10) and CSVs for the full per-condition numbers.